

Ebola, Sierra Leone, 2014

Overview

This dataset contains individual-level line list data from the 2014 Ebola outbreak in Sierra Leone. With 11,903 records, it includes patient demographics (age, sex), clinical status, dates of onset and sample collection, and geographic information (district and chiefdom). This is one of the largest publicly available Ebola line lists.

Source: Fang LQ, Yang Y, Jiang JF, Yao HW, Kargbo D, Li XL, et al. (2016). Transmission dynamics of Ebola virus disease and intervention effectiveness in Sierra Leone. *Proceedings of the National Academy of Sciences* 113(16):4488–4493.

Loading the data

```
using DataFrames
using Outbreaks
df = Outbreaks.ebola_sierraleone_2014()
size(df)
```

(11903, 8)

First few rows

```
first(df, 6)
```

	id	age	sex	status	date_of_onset	date_of_sample	district	chiefdom
	Int64	String7	String3	String15	Date	Date	String15	String31
1	1	20	F	confirmed	2014-05-18	2014-05-23	Kailahun	Kissi Teng
2	2	42	F	confirmed	2014-05-20	2014-05-25	Kailahun	Kissi Teng
3	3	45	F	confirmed	2014-05-20	2014-05-25	Kailahun	Kissi Tonge
4	4	15	F	confirmed	2014-05-21	2014-05-26	Kailahun	Kissi Teng
5	5	19	F	confirmed	2014-05-21	2014-05-26	Kailahun	Kissi Teng
6	6	55	F	confirmed	2014-05-21	2014-05-26	Kailahun	Kissi Teng

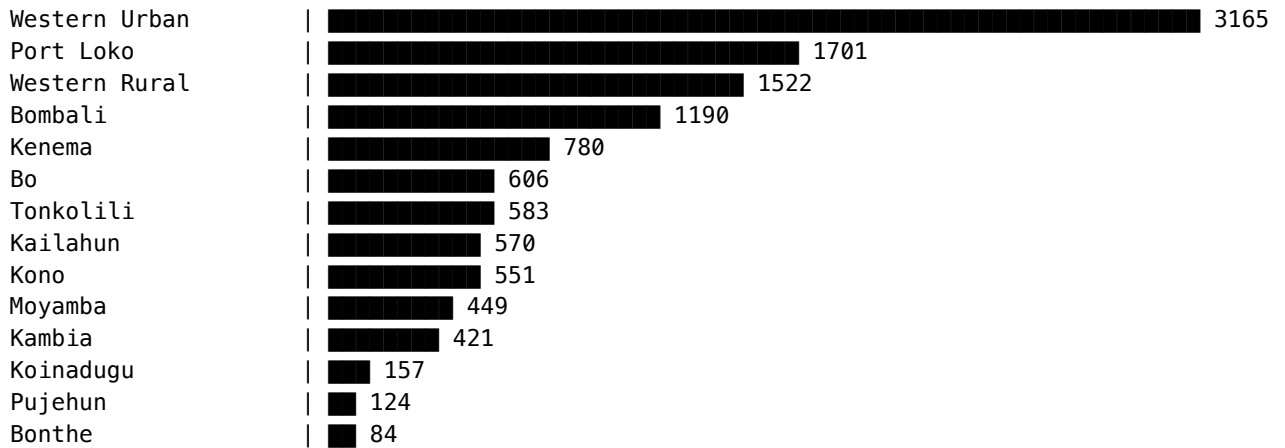
Summary statistics

```
describe(df)
```

	variable	mean	min	median	max	nmissing	eltype
	Symbol	Union...	Any	Any	Any	Int64	DataType
1	id	5952.0	1	5952.0	11903	0	Int64
2	age		0		NA	0	String7
3	sex		F		NA	0	String3
4	status		confirmed		suspected	0	String15
5	date_of_onset		2014-05-18	2014-11-18	2015-09-12	0	Date
6	date_of_sample		2014-05-23	2014-11-24	2015-09-13	0	Date
7	district		Bo		Western Urban	0	String15
8	chiefdom		Badjia		Yoni	0	String31

Cases by district

```
using StatsBase
district_counts = sort(combine(groupby(df, :district), nrow => :cases), :cases, rev=true)
for row in eachrow(district_counts)
    bar = repeat("█", max(1, round{Int, row.cases / 50}))
    println(rpad(coalesce(string(row.district), "missing"), 20), " | $bar $(row.cases)")
end
```



Epidemic curve (weekly)

```
onset = dropmissing(df, :date_of_onset)
sort!(onset, :date_of_onset)
println("Onset range: $(minimum(onset.date_of_onset)) to $(maximum(onset.date_of_onset))")
println("Cases with onset date: $(nrow(onset)) / $(nrow(df))")
```

Onset range: 2014-05-18 to 2015-09-12
Cases with onset date: 11903 / 11903